Precalculus Review for Calculus

ANSWER KEY

Functions and composition

1 Let $f(x) = \sqrt{x-1}$ and $g(x) = x^2 + 3$. Find:

a $(f \circ g)(x)$ $f(g(x)) = \sqrt{x^2+2}$.

b $(g \circ f)(x)$ $g(f(x)) = (x-1) + 3 = x + 2$, for $x \geq 1$.

c the domain of $f \circ g$ $x^2 + 2 \geq 1$ always holds, so the domain is all real numbers.

2 Simplify the difference quotient $\frac{f(x+h) - f(x)}{h}$ for $f(x) = x^2 - 3x$.
$\frac{(x+h)^2 - 3(x+h) - (x^2 - 3x)}{h} = \frac{2xh + h^2 - 3h}{h} = 2x + h - 3$.

Lines, exponentials, and trig — the toolkit calculus builds on

3 Find the slope and equation of the line through $(-1, 4)$ and $(3, -2)$.
Slope $= \frac{-2-4}{3-(-1)} = -\frac{3}{2}$; equation $y - 4 = -\frac{3}{2}(x+1)$, i.e. $y = -\frac{3}{2}x + \frac{5}{2}$.

4 Solve $2e^{3x} = 14$ exactly and to three decimal places.
$e^{3x} = 7$, so $x = \frac{\ln 7}{3} \approx 0.649$.

5 State the exact value of each expression, needed frequently when differentiating trig functions:

a $\sin\frac{\pi}{4}$ $= \frac{\sqrt{2}}{2}$ **b** $\cos\frac{\pi}{3}$ $= \frac{1}{2}$ **c** $\tan\frac{\pi}{6}$ $= \frac{\sqrt{3}}{3}$

Factoring and rational expressions

6 Factor each expression fully:

a $x^2 - 5x + 6$ $= (x-2)(x-3)$ **b** $4x^2 - 9$ $= (2x-3)(2x+3)$ **c** $x^3 - 8$ $= (x-2)(x^2+2x+4)$

7 Simplify $\frac{x^2-4}{x^2-x-6}$, stating the value(s) of x excluded from the domain.
$\frac{(x-2)(x+2)}{(x-3)(x+2)} = \frac{x-2}{x-3}$, excluding $x = -2$ (removable) and $x = 3$ (undefined).

Inverse functions and logarithms

8 Find the inverse of $f(x) = 3x - 5$, and verify by computing $f(f^{-1}(x))$.
Swap and solve: $x = 3y - 5 \Rightarrow y = \frac{x+5}{3}$, so $f^{-1}(x) = \frac{x+5}{3}$. Check:
$f(f^{-1}(x)) = 3 \cdot \frac{x+5}{3} - 5 = x + 5 - 5 = x$. ✓

9 Solve $\ln(x-1)=2$ exactly and to three decimal places.

$x-1=e^2$, so $x=e^2+1\approx 8.389$.

Piecewise and absolute value review

10 Sketch-reasoning: rewrite $f(x)=|2x-6|$ as a piecewise function without absolute value bars.

$f(x)=2x-6$ for $x\geq 3$, and $f(x)=6-2x$ for $x<3$ (since $2x-6\geq 0$ exactly when $x\geq 3$).

11 Solve the inequality $|3x+1|\leq 7$, writing your answer in interval notation.

$-7\leq 3x+1\leq 7$: $-8\leq 3x\leq 6$: $-\frac{8}{3}\leq x\leq 2$. Interval: $\left[-\frac{8}{3},2\right]$.

Systems of equations

12 Solve the system $2x+y=7$ and $x-y=2$ by elimination.

Adding the equations: $3x=9$, so $x=3$; then $y=7-2(3)=1$. Solution: $(3,1)$.

13 Find the point(s) of intersection of $y=x^2$ and $y=2x+3$.

$x^2=2x+3$: $x^2-2x-3=0$, $(x-3)(x+1)=0$: $x=3,\,-1$. Points: $(3,9)$ and $(-1,1)$.